

Pretty Species Info Table for Paper

Heywood et al.

```
library(tidyverse)

Species_info <- left_join(read_csv('../data/brood_table.csv'),
                          read_csv('../data/names_table.csv'))>%
  select( Sp = `Taxon name`,
          maximum_plausible,
          minimum_plausible)

NHM_ForFitting <- read_csv( '../data/NHM_for_models_monthly.csv')>%
  left_join(Species_info)
OUM_ForFitting <- read_csv( '../data/OUMNH_for_models_monthly.csv')
SpeciesToInclude<- unique(read_csv('../statsresults/NHM_Posterior_Summaries_M1.csv')$Sp)

UKRedList <- read_csv('../data/ButterflyRedlistUK.csv')
specialism <- read_csv( '../data/specialism_with_nhm.csv')
```

Making Table 1

Table summarising each species. SI Table 1

```
left_join(NHM_ForFitting %>% group_by(Sp) %>%
  count(name = 'NHM_n') ,
  OUM_ForFitting %>%group_by(Sp) %>%
  count(name = 'OUM_n') ) -> CountsUsedInModels

## Joining with 'by = join_by(Sp)'

left_join(read_csv('../data/brood_table.csv'),
          read_csv('../data/names_table.csv'))>%
  left_join(specialism) %>%
  filter(`Taxon name` %in% SpeciesToInclude) %>%
  left_join(UKRedList, by = c('Taxon name'='Species')) %>%
  left_join(CountsUsedInModels, by = c('Taxon name' = 'Sp')) %>%
  mutate( Habitat = ifelse( Specialism %in% c('Habitat and foodplant generalist' ,
                                              'Habitat generalist and foodplant specialist'),
                           'G', 'S'),
          Foodplant = ifelse( Specialism %in% c('Habitat and foodplant generalist' ,
                                              'Habitat specialist and foodplant generalist'),
                              'G', 'S')) %>%
  mutate( `Focal Brood` = ifelse(Univoltine,
```

```

        'Univoltine',
        paste0( FocalBrood, ': ',
                month.abb[FirstMonthBrood_focal], '-',
                month.abb[LastMonthBrood_focal])) %>%
arrange(Univoltine, `Taxon name` ) %>%
mutate(`Common Name` = ifelse( !is.na(EnglishName), EnglishName,
                              ifelse( !is.na(`Elsa name` ),
                                      `Elsa name` ,
                                      'TO BE ADDED') )) %>%
mutate(`GB Red List category` = ifelse( is.na(`GB Red List category`),
                                         'Not Listed',
                                         `GB Red List category` ) %>%

select(Species = `Taxon name`,
       `Common Name`,
       Family,
       Habitat,
       Foodplant,
       NHM_n,
       OUM_n,
       `Focal Brood`,
       FocalMonthLateLarval ,
       `GB Red List category` ) %>%
knitr::kable()

```

```

## Rows: 100 Columns: 19
## -- Column specification -----
## Delimiter: ", "
## chr (12): Elsa name, Taxon name, NHM name, Family, Month egg laid, Month la...
## dbl (5): FocalBrood, FirstMonthBrood_focal, LastMonthBrood_focal, maximum_p...
## lgl (2): Include, Univoltine
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
## Rows: 55 Columns: 3
## -- Column specification -----
## Delimiter: ", "
## chr (3): Elsa name, NHM name, Taxon name
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
## Joining with 'by = join_by('Elsa name', 'Taxon name', 'NHM name')'
## Joining with 'by = join_by('Taxon name')'

```

Species	Common Name	Family	Habitat	Foodplant	NHM_n	OUM_n	Focal Brood	FocalMonthLateLarval	GB Red List category
Aglaia urticae	Small tortoiseshell	Nymphalidae	C	S	971	34	1: Jun-Aug	May	Not Listed
Aricia agestis	Brown argus	Lycaenidae	S	S	975	42	2: Aug-Oct	July	Not Listed

Species	Common Name	Family	Habitat	Foodplant	NHM	QUM	Focal Brood	FocalMonth	GB Red List Category
Celastrina argiolus	Holly blue	Lycaenidae	G	G	303	25	2: Aug-Oct	June	Not Listed
Lasiommata megera	Wall	Nymphalidae	Gae	G	619	62	2: Aug-Oct	June	Endangered
Leptidea sinapis	Wood White	Pieridae	S	S	791	70	1: Mar-Jun	JulyBefore	Endangered
Lycaena phlaeas	Small copper	Lycaenidae	G	S	2877	65	2: Jul-Oct	June	Not Listed
Pararge aegeria	Speckled wood	Nymphalidae	Gae	G	471	38	2: Jul-Aug	June	Not Listed
Pieris brassicae	Large white	Pieridae	G	S	344	44	2: Jul-Sep	June	Not Listed
Pieris napi	Green-veined white	Pieridae	G	G	766	NA	2: Jul-Sep	June	Not Listed
Pieris rapae	Cabbage white	Pieridae	G	G	445	NA	2: Jul-Sep	June	Not Listed
Polygonia c-album	Comma	Nymphalidae	Gae	G	466	39	1: Jun-Aug	June	Not Listed
Polyommatus bellargus	Adonis Blue	Lycaenidae	S	S	4830	63	2: Aug-Sep	July	Vulnerable
Polyommatus icarus	Common blue	Lycaenidae	G	G	3752	63	2: Aug-Sep	July	Not Listed
Aglais io	Peacock	Nymphalidae	Gae	S	505	80	Univoltin	June	Not Listed
Anthocharis cardamines	Orange tip	Pieridae	G	G	1661	64	Univoltin	JulyBefore	Not Listed
Apatura iris	Purple emperor	Nymphalidae	Sae	S	124	NA	Univoltin	March	Not Listed
Aphantopus hyperantus	Ringlet	Nymphalidae	Gae	G	2206	NA	Univoltin	June	Not Listed
Argynnis paphia	Silver washed fritillary	Nymphalidae	Gae	S	1447	83	Univoltin	June	Not Listed
Boloria euphrosyne	Pearl-bordered Fritillary	Nymphalidae	Gae	S	1865	NA	Univoltin	April	Vulnerable
Boloria selene	Small Pearl-bordered Fritillary	Nymphalidae	Sae	S	1925	NA	Univoltin	April	Vulnerable
Callophrys rubi	Green hairstreak	Lycaenidae	G	G	990	NA	Univoltin	JuneBefore	Not Listed
Carterocephalus palaemon	Chequered Skipper	Hesperiidae	S	S	604	NA	Univoltin	AugustBefore	Not Listed
Coenonympha pamphilus	Small Heath	Nymphalidae	Gae	S	3058	NA	Univoltin	April	Vulnerable
Coenonympha tullia	Large Heath	Nymphalidae	Sae	S	1409	44	Univoltin	April	Endangered

Species	Common Name	Family	Habitat	Foodplant	NHM	QUM	Focal Brood	Focal Month	Local Lagony	GB Red List
Cupido minimus	Small Blue	Lycaenidae	G	S	1453	89	Univoltine	July	Before	Near Threatened
Erynnis tages	Dingy skipper	Hesperiidae	G	S	856	85	Univoltine	August	Before	Not Listed
Euphydryas aurinia	Marsh Fritillary	Nymphalidae	Gae	S	1926	NA	Univoltine	May		Vulnerable
Fabriciana adippe	High Brown Fritillary	Nymphalidae	Gae	S	994	NA	Univoltine	May		Endangered
Favonius quercus	Purple hairstreak	Lycaenidae	G	S	494	83	Univoltine	May		Not Listed
Gonepteryx rhamni	Brimstone	Pieridae	G	S	483	81	Univoltine	June		Not Listed
Hamearis lucina	Duke of Burgundy	Riodinidae	S	S	951	NA	Univoltine	July	Before	Vulnerable
Hesperia comma	Silver-spotted Skipper	Hesperiidae	S	S	646	103	Univoltine	June		Vulnerable
Hipparchia semele	Grayling	Nymphalidae	Sae	G	1654	NA	Univoltine	April		Endangered
Limenitis camilla	White Admiral	Nymphalidae	Sae	S	761	97	Univoltine	May		Vulnerable
Maniola jurtina	Meadow brown	Nymphalidae	Gae	G	4730	70	Univoltine	April		Not Listed
Melanargia galathea	Marbled white	Nymphalidae	Gae	G	2169	101	Univoltine	May		Not Listed
Melitaea athalia	Heath Fritillary	Nymphalidae	Gae	G	1619	NA	Univoltine	March		Endangered
Melitaea cinxia	Glanville Fritillary	Nymphalidae	Sae	S	579	NA	Univoltine	March		Endangered
Nymphalis polychloros	Large Tortoiseshell	Nymphalidae	Gae	G	197	NA	Univoltine	June		Regionally Extinct
Ochlodes sylvanus	Large skipper	Hesperiidae	G	G	1029	NA	Univoltine	May		Not Listed
Papilio machaon	Swallowtail	Papilionidae	Se	S	166	54	Univoltine	July	Before	Vulnerable
Plebejus argus	Silver-studded Blue	Lycaenidae	S	G	7105	NA	Univoltine	May		Vulnerable
Polyommatus coridon	Chalk Hill Blue	Lycaenidae	G	S	19584	NA	Univoltine	June		Vulnerable
Pyrgus malvae	Grizzled Skipper	Hesperiidae	G	G	1285	108	Univoltine	July		Vulnerable
Pyronia tithonus	Gatekeeper	Nymphalidae	Gae	S	2313	125	Univoltine	June		Not Listed
Satyrium pruni	Black Hairstreak	Lycaenidae	S	S	377	NA	Univoltine	May		Endangered
Satyrium w-album	White-letter Hairstreak	Lycaenidae	S	S	346	NA	Univoltine	May		Vulnerable
Speyeria aglaja	Dark Green Fritillary	Nymphalidae	Gae	S	1028	51	Univoltine	May		Near Threatened
Thecla betulae	Brown Hairstreak	Lycaenidae	G	S	165	NA	Univoltine	June		Vulnerable

Species	Common Name	Family	Habitat	Foodplant	NHM	QUM	Brood	FocalMonth	GB Red List Category
Thymelicus acteon	Lulworth Skipper	Hesperiidae	S	S	544	NA	Univoltin	May	Near Threatened
Thymelicus lineola	Essex skipper	Hesperiidae	G	G	707	NA	Univoltin	May	Not Listed
Thymelicus sylvestris	Small skipper	Hesperiidae	G	G	918	98	Univoltin	May	Not Listed

Making Table 2

Table of Year Coefficients vs Red listing

```
PS_NHM<- read_csv('../statsresults/NHM_Posterior_Summaries_Spatial.csv')

## Rows: 9947 Columns: 10
## -- Column specification -----
## Delimiter: ","
## chr (2): variable, Sp
## dbl (8): mean, median, sd, mad, q5, q95, q16.7, q83.3
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.

UKRedList <- read_csv('../data/ButterflyRedlistUK.csv')

## Rows: 33 Columns: 3
## -- Column specification -----
## Delimiter: ","
## chr (3): EnglishName, Species, GB Red List category
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.

PS_NHM %>%
  left_join(UKRedList, by =c( 'Sp'='Species') ) %>%
  mutate(RedListListed = ifelse(is.na( `GB Red List category` ),
                                'Unlisted',
                                'Listed')) %>%

  filter( variable == 'b_Temp' ) %>%
  arrange( RedListListed,mean)%>%
  select(Species = Sp, RedListing = RedListListed,
         `GB Red List category`, mean ,q5,q95) %>%
  knitr::kable(digits = 4)
```

Species	RedListing	GB Red List category	mean	q5	q95
<i>Thecla betulae</i>	Listed	Vulnerable	-0.0080	-0.0148	-0.0011
<i>Nymphalis polychloros</i>	Listed	Regionally Extinct	-0.0066	-0.0177	0.0046
<i>Leptidea sinapis</i>	Listed	Endangered	-0.0045	-0.0072	-0.0020
<i>Coenonympha tullia</i>	Listed	Endangered	-0.0038	-0.0058	-0.0017
<i>Pyrgus malvae</i>	Listed	Vulnerable	-0.0033	-0.0052	-0.0012
<i>Hamearis lucina</i>	Listed	Vulnerable	-0.0019	-0.0040	0.0001
<i>Boloria selene</i>	Listed	Vulnerable	-0.0010	-0.0034	0.0013
<i>Polyommatus bellargus</i>	Listed	Vulnerable	-0.0009	-0.0018	0.0001
<i>Cupido minimus</i>	Listed	Near Threatened	-0.0005	-0.0033	0.0023
<i>Satyrrium pruni</i>	Listed	Endangered	-0.0002	-0.0035	0.0030
<i>Polyommatus coridon</i>	Listed	Vulnerable	0.0005	-0.0003	0.0014
<i>Coenonympha pamphilus</i>	Listed	Vulnerable	0.0009	-0.0009	0.0028
<i>Boloria euphrosyne</i>	Listed	Vulnerable	0.0011	-0.0007	0.0028
<i>Lasiommata megera</i>	Listed	Endangered	0.0011	-0.0023	0.0045
<i>Plebejus argus</i>	Listed	Vulnerable	0.0013	-0.0003	0.0029
<i>Papilio machaon</i>	Listed	Vulnerable	0.0018	-0.0061	0.0098
<i>Satyrrium w-album</i>	Listed	Vulnerable	0.0029	-0.0024	0.0081
<i>Hipparchia semele</i>	Listed	Endangered	0.0030	0.0001	0.0059
<i>Melitaea athalia</i>	Listed	Endangered	0.0040	0.0019	0.0062
<i>Limenitis camilla</i>	Listed	Vulnerable	0.0048	0.0017	0.0078
<i>Melitaea cinxia</i>	Listed	Endangered	0.0049	0.0009	0.0091
<i>Euphydryas aurinia</i>	Listed	Vulnerable	0.0050	0.0014	0.0085
<i>Speyeria aglaja</i>	Listed	Near Threatened	0.0051	0.0024	0.0077
<i>Thymelicus acteon</i>	Listed	Near Threatened	0.0076	0.0029	0.0123
<i>Hesperia comma</i>	Listed	Vulnerable	0.0088	0.0054	0.0122
<i>Fabriciana adippe</i>	Listed	Endangered	0.0108	0.0083	0.0134
<i>Apatura iris</i>	Unlisted	NA	-0.0141	-0.0207	-0.0076
<i>Celastrina argiolus</i>	Unlisted	NA	-0.0103	-0.0179	-0.0028
<i>Polygonia c-album</i>	Unlisted	NA	-0.0059	-0.0099	-0.0018
<i>Pararge aegeria</i>	Unlisted	NA	-0.0049	-0.0087	-0.0012
<i>Pieris rapae</i>	Unlisted	NA	-0.0045	-0.0086	-0.0004
<i>Argynnis paphia</i>	Unlisted	NA	-0.0041	-0.0066	-0.0017
<i>Thymelicus lineola</i>	Unlisted	NA	-0.0035	-0.0070	0.0000
<i>Gonepteryx rhamni</i>	Unlisted	NA	-0.0034	-0.0065	-0.0003
<i>Aricia agestis</i>	Unlisted	NA	-0.0031	-0.0061	-0.0002
<i>Callophrys rubi</i>	Unlisted	NA	-0.0030	-0.0063	0.0003
<i>Aphantopus hyperantus</i>	Unlisted	NA	-0.0028	-0.0051	-0.0005
<i>Anthocharis cardamines</i>	Unlisted	NA	-0.0020	-0.0045	0.0006
<i>Pieris napi</i>	Unlisted	NA	-0.0013	-0.0042	0.0016
<i>Lycaena phlaeas</i>	Unlisted	NA	-0.0010	-0.0033	0.0013
<i>Melanargia galathea</i>	Unlisted	NA	-0.0008	-0.0037	0.0022
<i>Aglais urticae</i>	Unlisted	NA	-0.0006	-0.0047	0.0034
<i>Maniola jurtina</i>	Unlisted	NA	0.0005	-0.0011	0.0022
<i>Thymelicus sylvestris</i>	Unlisted	NA	0.0006	-0.0019	0.0031
<i>Carterocephalus palaemon</i>	Unlisted	NA	0.0008	-0.0022	0.0038
<i>Pyronia tithonus</i>	Unlisted	NA	0.0010	-0.0009	0.0029
<i>Polyommatus icarus</i>	Unlisted	NA	0.0015	0.0000	0.0030
<i>Favonius quercus</i>	Unlisted	NA	0.0018	-0.0026	0.0063
<i>Aglais io</i>	Unlisted	NA	0.0027	-0.0041	0.0095
<i>Erynnis tages</i>	Unlisted	NA	0.0033	0.0002	0.0064
<i>Ochlodes sylvanus</i>	Unlisted	NA	0.0037	0.0011	0.0064
<i>Pieris brassicae</i>	Unlisted	NA	0.0043	0.0001	0.0086